

Sensor applications

Photoacoustic Time–Frequency Analysis for Quantification of Coal Thermal Properties and its Classification to Spontaneous Combustion Susceptibility

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Abstract—Accurate quantitative assessment of coal thermal properties is critical for predicting spontaneous combustion susceptibility (SCS), particularly for on site evaluation in coal stockyards where early decision making can mitigate unwanted coal burning and greenhouse gas emissions. This work presents A.S.P.E.C.T. (AI-Supported Photoacoustic Evaluator of Coal Thermodynamics), a compact, pulsed laser diode–based photoacoustic (PA) sensing system for on-field assessment of coal ignition temperature. The novelty of the proposed method lies in the exploitation of a time–frequency-based PA feature set, derived through a dedicated signal processing framework and enhanced interpretability of the PA data through Wigner-Ville distribution-based spectrograms, enabling ignition temperature quantification with an accuracy of approximately 89%. In addition, the propensity of coal toward spontaneous combustion is classified into high, moderate, and low SCS categories using convolutional neural network classification, achieving an average accuracy of 93%. The developed system is also applied to raw coal samples towards their thermal property assessment and predicting its SCS. The system is integrated with user interface and can evaluate a coal sample in less than 1 minute.

Index Terms— Coal spontaneous combustion susceptibility, photoacoustic sensing, time-frequency analysis, Wigner-Ville spectrograms for coal classification.

I. INTRODUCTION

Coal is a vital energy resource and industrial raw material, making its safe storage a continuing concern [1]. In stockyards, self-heating can lead to spontaneous combustion, causing fires, hazardous gas emissions, and significant economic and safety impacts [2]. Therefore, reliable assessment of coal thermal properties is essential for evaluating heat generation and retention.

The spontaneous combustion of coal originates from low-temperature oxidation reactions occurring at active surface sites. Intrinsic characteristics such as coal rank, pore structure, oxygen-containing functional groups, volatile matter content, moisture behavior, and oxygen adsorption capacity influence oxidation kinetics and heat accumulation. Coals possessing highly developed pore networks and greater concentrations of reactive functional groups generally exhibit lower ignition temperatures and higher susceptibility to spontaneous combustion [3]. Coal self-heating tendency has traditionally been evaluated using offline thermal-response techniques such as thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), and crossing point temperature (CPT) measurements [4]. More recently, these indicators have been combined with data-driven models to classify combustion risk and support mining operations [5]. However, these approaches provide only static assessments and cannot capture the evolving thermal state of coal in stockyards. Existing field methods, including gas monitoring and chemical or moisture-based treatments, either detect risk after combustion initiation or may affect coal quality [6]. Because coal stockpiles are dynamic systems influenced by environmental conditions, pile geometry, and material heterogeneity, periodic or offline assessments are inadequate for reliable risk evaluation [7]. Since spontaneous combustion arises from an

imbalance between heat generation and dissipation, continuous monitoring of coal thermal behavior is essential for assessing environmental effects and mitigating storage-related combustion risk. This paper presents a field-deployable, AI-integrated photoacoustic (PA) sensing framework for early prediction of coal SCS. PA sensing uses short laser pulses to generate thermoelastic acoustic waves governed by coal thermophysical properties, enabling rapid, non-destructive assessment of thermal behavior [8,9]. Features extracted from time- and frequency-domain PA signals provide an effective basis for SCS prediction [10].

The novelty lies in a physically grounded and quantitatively validated PA-AI framework that incorporates (i) thermophysical property-linked multi-domain features, (ii) ignition temperature (T_{ign}) prediction, (iii) machine learning and Wigner–Ville spectrogram-based CNN models for robust classification, (iv) a compact field-deployable system, and (v) validation on raw coal samples. This bridges laboratory feasibility and practical coal stockyard deployment.

II. MATERIALS AND METHODS

A. Sample preparation & Ground truth measurement

Coal pellets for uniform photoacoustic measurements were prepared by crushing, grinding, and sieving raw coal to 212 μm , followed by homogeneous mixing with an epoxy binder (2:1 resin:hardener) at equal binder-to-coal mass. The mixture was molded into pellets of 40 mm diameter and 6 mm thickness and cured for ~ 90 min. TGA-derived ignition temperature (T_{ign}) served as the ground truth to classify 16 coal samples into low, moderate, and high SCS categories using K-means clustering. About 8 pellets per sample were prepared, and PA measurements from 5–6 positions per pellet produced 700 instances after outlier removal. Additionally, raw coal samples (3×3 cm, 5 mm thick) were tested using the developed system. Coal samples were collected from mines across India, including Nagpur, Dhanbad, and Talcher. Sample distribution is summarized in Table 1,

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while detailed pellet preparation and TGA procedures are given in [11]. Also, Table 2 summarizes the average proximate analysis parameters for each coal class, including moisture, ash content (A.C.), volatile matter (V.M.), and fixed carbon (F.C.), which strongly influence coal oxidation and self-heating behavior. In general, higher A.C. suppresses spontaneous combustion by reducing reactive combustible matter and increasing thermal inertia, whereas higher V.M. enhances oxidation reactivity. Moisture affects heat and oxygen transfer, while increased F.C. content is generally associated with higher ignition temperatures and lower spontaneous combustion susceptibility [12,13].

Table 1: K-means clustering of coal samples

Coal Samples	Ignition Temperature (T _{ign})	Centroid T _{ign} /Classes
C1,C2,C3,C4,C5,C6	(465, 476.5, 412.3, 439.7, 417.8, 440.3)°C	440°C/ Low
C7,C8,C9,C10,C11	(346.8, 355, 345.7, 353.7, 356.8)°C	360°C/Moderate
C12,C13,C14,C15,C16	(319.1, 334.6, 332.5, 333.7, 339.8)°C	323°C/High

Table 2: Average values of proximity analysis parameters for each class of coal

Sample Class towards SCS	Moisture (%)	F.C. (%)	A.C. (%)	V.M. (%)	T _{ign} . (°C)
Low	1.33	39.58	32.42	20.67	476.5
Moderate	6.83	34.84	20.1	27.33	346.8
High	8.01	28.43	13.09	34.16	319.2

B. Experimental setup

A compact, low-power PA sensing platform was developed for portable on-field coal assessment. Optical excitation was provided by a 905 nm pulsed laser diode (400 ns pulse width, 10 μJ pulse energy), focused to a 1 mm spot and operated at 1 kHz. PA signals were detected using a 1 MHz piezoelectric ultrasound (V302-SU, Olympus Inc. USA) sensor coupled to the coal pellet, as shown in Fig. 1(a). Owing to the weak and heterogeneous acoustic response of coal, signals were amplified using an in-house low-noise preamplifier (40 dB gain, 5 MHz bandwidth) and averaged over 100 laser shots to improve SNR. Data was acquired at a sampling rate of 50 MHz through a hardware-synchronized DAQ system for repeatable time-resolved measurements suitable for portable stockyard monitoring. Fig. 1(b) presents the field-deployable A.S.P.E.C.T. system with all hardware integrated into a custom 3D-printed enclosure, and Fig. 1(c) shows the Python-based user interface for ignition temperature prediction and SCS classification.

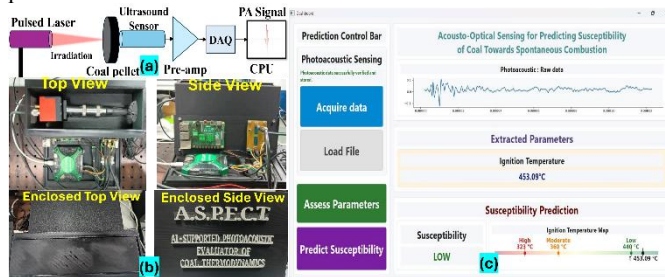


Fig. 1: Experimental setup: (a) Schematic, (b).System design and (c).UI

C. Signal Processing Framework

Fig. 2 illustrates the PA signal-processing framework for extracting thermally relevant features to evaluate coal thermal properties and predict SCS. Raw PA signals are analyzed using FFT, while Savitzky-Golay smoothing and wavelet denoising suppress noise while preserving thermoelastic information. Time-domain (peak amplitude, RMS, skewness), frequency-domain (spectral centroid, bandwidth, entropy, dominant frequency), wavelet-domain energy/entropy, and SNR features capture absorbed energy, thermal relaxation, diffusivity, heterogeneity, and non-stationary heat transport, serving as inputs for regression and classification models for coal thermal assessment and SCS categorization.

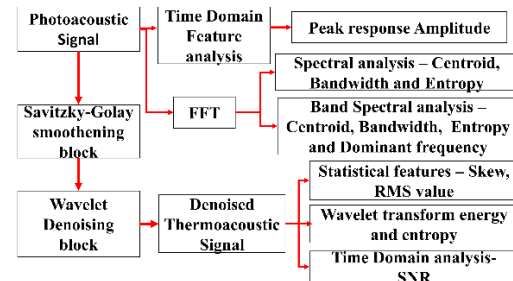


Fig. 2: Signal Processing Framework

III. RESULTS AND DISCUSSIONS

Fig. 3(a,b) shows a representative PA signal and its corresponding frequency spectrum taken from a coal pellet. The time-domain signals exhibit class-dependent variations in peak amplitude and decay, reflecting differences in thermoelastic response and heat dissipation among coal samples. A stable baseline offset arises from DC bias and low-frequency thermoelastic background. The spectra indicate that dominant energy lies below ~1–1.2 MHz, with subtle class-specific amplitude variations within this range, associated with acoustic attenuation and internal damping rather than bandwidth changes. Figure 4(a,b) presents the PA signal and the corresponding spectra of a coal pellet and a raw coal sample collected from the same mine. The observed signal shift is likely attributable to the greater surface roughness and inherent inhomogeneity of the raw coal sample compared to the relatively smooth and homogeneous pelletized sample. The PA response depends on the thermal diffusion length (μ)-

$$\mu = \sqrt{\frac{\alpha}{\pi f}}$$

where α is thermal diffusivity and f is modulation frequency. Samples with different diffusivities exhibit distinct frequency-domain attenuation and spectral characteristics [14]. Moreover, the thermal relaxation time (τ) governs the temporal decay of PA signals and shorter relaxation times result in faster decay, reflecting efficient heat dissipation [15]. Further, oxidation behavior (linked with ignition temperature) influences heat generation rate, internal damping, acoustic attenuation and microstructural heterogeneity. Therefore, this study manifests these effects in multi-domain features as the time-domain (amplitude, rise/decay) reflecting thermal relaxation and energy absorption; frequency-domain (centroid, bandwidth); governed by thermal diffusivity and effusivity [14], whereas Time-frequency (Wigner-Ville distribution): captures non-stationary thermal processes and evolving oxidation behavior [16]. Therefore, low-frequency spectral descriptors were used instead of peak frequency alone. Initially, time- and frequency-domain features were used to develop a T_{ign} regression model with an 80:20 train-test split,

yielding 82% cross-validation and 85.41% test accuracy. Savitzky-Golay smoothing and wavelet-based denoising were applied before to improve noise suppression and thermoelastic signal fidelity.

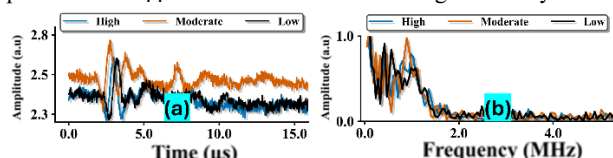


Fig 3: (a). PA signals and (b). corresponding spectra- for low, moderate, and high SCS

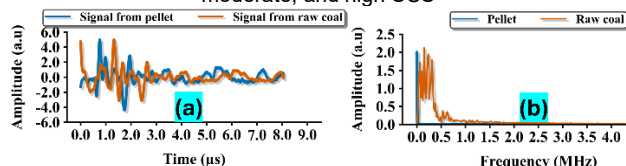


Fig 4: (a). PA signals & (b). spectra- for raw coal and pellet

Fig. 5 compares training and testing accuracies of regression models using the same feature sets under three conditions: (a) raw features (R) from raw time-domain signals and FFT spectra, (b) partially processed features (PP), with processed time-domain features and frequency-domain features from raw-signal FFT, and (c) processed features (P), with both time- and frequency-domain features extracted from fully processed signals.



Fig.5: Bar chart representing regression model accuracy

As shown in Fig. 5, PP (b) achieved the highest training and testing accuracy and was therefore selected for ignition temperature assessment. Fig. 6(a) compares predicted and true ignition temperatures, showing a strong linear relationship with data closely aligned to the ground-truth line, indicating effective capture of the relationship between photoacoustic features and ignition temperature. The model achieved high predictive performance ($R^2 \approx 85.4\%$, RMSE ≈ 5.7 , MAE ≈ 4.7) with minimal bias. Slightly higher dispersion at elevated temperatures is likely due to increased complexity in thermal decomposition and acoustic signal propagation [17]. Fig. 6(b,c) further validate the model through Q-Q residual analysis and Bland-Altman agreement analysis.

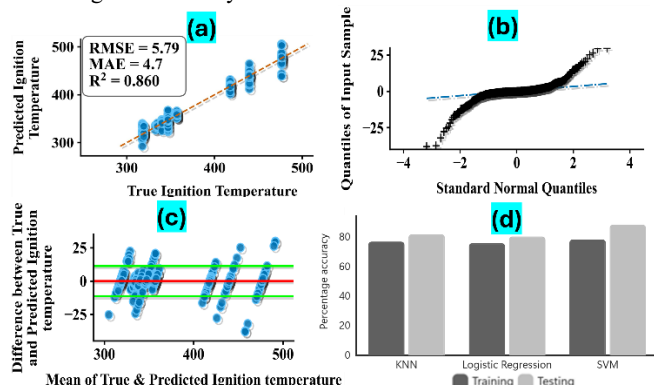


Fig.6: (a)True Vs predicted ignition temperature of coal, (b) Q-Q plot, (c) Bland Altman plot and (d) Classification accuracy for three models

The regression model's validity is confirmed by residual analysis, with the Q-Q plot (Fig. 6(b)) showing near-Gaussian residuals and the Bland-Altman plot (Fig. 6(c)) indicating near-zero bias and weak heteroscedasticity, demonstrating uniform performance across the prediction range [18]. Using the PP feature set and by removing the outliers (± 1.96 standard deviation), the accuracy to measure ignition temperature has increased the test R^2 from 81.54% to 85.41% ($\sim 86\%$) and reduced the RMSE from 15.17% to 5.79%, indicating that extreme observations introduced significant variability in the model.

The same feature set was used for three-class coal classification (low, moderate, and high SCS; Table 1) with 240, 220, and 240 instances, respectively. KNN outperformed SVM and logistic regression (Fig. 6-d) by capturing local similarities in the PA feature space, whereas the others rely on global decision boundaries. KNN used $k=5$ and Euclidean distance; Logistic Regression used a linear ECOC framework; Bagged Trees used 200-300 trees with a maximum of 10 splits; and SVM used an RBF kernel with ECOC.

To improve accuracy and interpretability, Wigner-Ville spectrogram (WVS)-based CNN classification was employed instead of feature-based methods. The WVS-CNN transforms one-dimensional PA signals into two-dimensional time-frequency maps, enabling automatic feature learning and visualization of class-specific patterns (Fig. 7). CNN used spectrogram inputs, three convolutional layers (32-128 filters, 3×3 kernels), ReLU activation, max-pooling, Adam optimization (learning rate = 0.001, batch size = 32), and a Softmax output layer for 3-class classification.

For the full dataset, classwise WVS, CNN outputs, and Grad-CAM attention maps were obtained. In Fig. 7 (HA1, MA1, and LA1), the attention maps represent the intersection of all attention layers within each class. Using an 80:20 pellet-level train-test split, the model achieved 91% cross-validation accuracy and nearly 88% test accuracy.

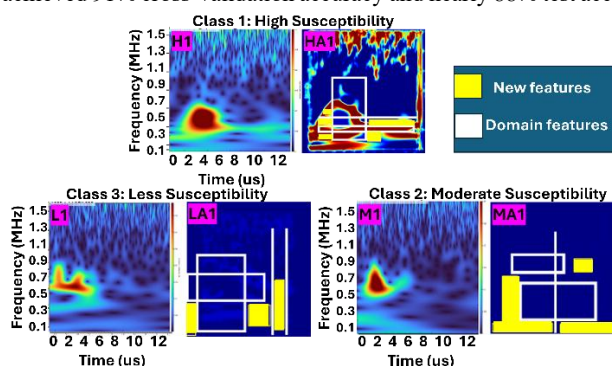


Fig.7: WVS (H1, M1, L1) and its corresponding attention layers (HA1, MA1, LA1) for three classes of coal SCS

Although feature-based classifiers (KNN, SVM, and logistic regression) effectively capture global feature-space trends, WVS-CNN provides complementary advantages by directly learning from high-resolution time-frequency representations, avoiding information loss from handcrafted features and capturing localized, non-stationary PA signal patterns. Grad-CAM maps (Fig. 7) enhance interpretability, where white rectangles largely correspond to handcrafted PP features and yellow rectangles identify additional discriminative regions associated with spectral entropy, spectral centroid, spectral bandwidth, and dominant-frequency amplitude. These frequency bands reflect different thermal diffusion and heat transport regimes, whose oxidation-induced changes indicate variations in thermal diffusivity and thermoelastic response related to self-heating and SCS. Consequently, these complementary representations improve both interpretability and regression/classification performance when combined with PP features (Table 3).

Table 3: Regression and classification accuracy with PP feature set and PP + WVS-CNN feature set

Feature set	PP		PP + (WVS-CNN)	
	Train (CV) accuracy	Test accuracy	Train (CV) accuracy	Test accuracy
Regression	82%	85.41%	~92%	89%
Classification	91%	88%	96%	93%

Figure 8 shows the confusion matrix of the coal SCS classification obtained through the combined features set of PP and WVS-CNN. In total, 130 of 139 samples (~ 20% test dataset of 700 samples) are correctly classified, corresponding to an overall accuracy of ~93%. Most errors occur in the moderate class due to overlapping features and limited data, but misclassifications are largely confined to adjacent classes, indicating stable and unbiased model performance.

True Class		Predicted Class			Class	Precision	Recall (Sensitivity)	F1-Score	Specificity
		High	Moderate	Less					
	High	44	2	2					
	Moderate	1	41	1					
	Less	2	1	45	Moderate	0.936	0.917	0.927	0.956
						0.932	0.953	0.943	0.948
		High	Moderate	Less	Low	0.938	0.938	0.938	0.956

Fig.8: Confusion matrix and metrics for three classes of coal

The confusion matrix and evaluation metrics show sparse, uniformly distributed errors, suggesting that misclassifications arise mainly from intrinsic class similarities rather than classifier limitations. The balanced performance across all classes demonstrates strong generalization, low bias, and effective learning of class-specific patterns, confirming the model's robustness for reliable multiclass decision-making. Using the same model on 24 PA data instances from 8 raw coal samples collected through the developed system yielded approximately 86% regression accuracy (test) and 92% classification accuracy.

While this study demonstrates high accuracy in ignition temperature estimation and SCS-based coal classification, further improvement may be achieved by incorporating additional coal parameters, such as fixed carbon and ash content, potentially inferred from the PA dataset. Although ambient temperature, humidity, airborne dust, and sample heterogeneity can influence PA measurements, their effects would be mitigated by using proposed custom 3D-printed enclosure, controlled sample preparation with polished surfaces, signal averaging, and multi-location measurements. Future studies will investigate environmental compensation and adaptive calibration methods to further improve field-deployment reliability. Moreover, extending the approach to a fully non-contact framework through optical interrogation of acoustic vibrations could enhance its practical applicability and industrial relevance for coal thermal property assessment and SCS classification.

IV. CONCLUSION

This study presents a compact, AI-enabled PA sensing framework for estimating coal ignition temperature and SCS. Using processed PA signals and time–frequency features from handcrafted methods and WVS-CNN, it achieves ~89% accuracy in ignition temperature estimation and 93% accuracy in coal classification. While primarily validated on coal pellets, results also demonstrate applicability to raw coal samples. The system performs both tasks in under a minute per sample and employs low-power, lightweight components, supporting portable, non-contact field deployment.

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